MILITARY
SPECIFICATION

MIL-D-12852A(SigC)
8 May 1958
Superseding
Specification MIL-D-12852(SigC)
30 June 1953

DIAL, TELEPHONE
(DIAL TA-45( )/GT)

1. SCOPE

1.1 Dial TA-45( )/GT is a telephone dial used to obtain automatic selection of individual telephone stations when used with automatic telephone central office equipment. It has a finger plate with 10 holes. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-T-76	Tape; Pressure, Sensitive Adhesive, Paper, Water-Resistant
QQ-S-571	Solder; Soft (Tin, Tin-Lead, Lead-Silver)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-635	Boxes, Fiber
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-591	Boxes; Fiberboard, Wood-Cleated (For Domestic Shipment)
PPP-B-601	Boxes; Wood-Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
UU-T-111	Tape; Paper,Gummed (Sealing and Securing)
VV-L-820	Lubricating Oil, Petroleum, Light

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JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier-Material; Greaseproofed, Flexible, (Waterproofed)
MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment: General Process for
MIL-P-3115	Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic-Resin

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MIL-L-3918	Lubricating oil, instrument, jewel bearing, non-spreading, low temperature
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes (For Ground Signal Equipment)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments
MIL-STD-169	Standard Extreme Temperature Cycle
MIL-STD-170	Moisture-Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for

DRAWINGS

SIGNAL CORPS

SC-DL-24467	Dial TA-45( )/GT
SC-C-33073	Form to be Used for Pilot Pack Process Sheet

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

a. For rail (both shall be used).

(1) Uniform Freight Classification No.4

George H. Dumas, Chairman
Uniform Freight Classification
202 Union Station
Chicago 6, Illinois

(2) Consolidated Freight Classification Rules No. 21

W. S. Flint, Chairman
Consolidated Classification Committee
1 Park Avenue and 33rd Street
New York 16, New York

b. For truck

National Motor Freight Classification No. A-2 and No. 14

F. G. Freund, Agent
American Trucking Association, Incorporated
1424 Sixteenth Street
Washington 6, D. C.

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3. REQUIREMENTS

3.1 Construction.- Dial, Telephone (Dial TA-45( )/GT) shall be constructed in accordance with Drawing SC-DL-24467 and the requirements of this specification. In case of conflict between the specification and drawings, the specification shall govern.

3.2 Soldering.-

3.2.1 Solder.- Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571.

3.2.2 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.2.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area, or resulting damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.4 Interchangeability.- Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. (See 4.7). When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable , the contracting officer shall be consulted.

3.5 Marking.-

3.5.1 General.- Marking shall conform to Specification MIL-M-13231.

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3.6 Tropicalization.-

3.6.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152, unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.

(2) It does not incorporate any material listed in 3.6.3 unless the material has been given the treatment specified herein or a treatment approved by the contracing officen

(3) It does not comply with (1) or (2) above, but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.6.2 Treatment of materials.-

3.6.2.1 Treating materials.- Treating materials containing a mercury bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.6.2.2 Toxicity.- Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.6.2.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.6.2.4 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.6.3 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.

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3.6.3 Materials not resistant to moisture and fungi.- (Cont'd)

(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.7 Service conditions.- The equipment shall meet the following service conditions:

3.7.1 Equipment operating.- The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) Operation.- Satisfactory operation for at least 500,000 dialing operations. Compliance shall be determined as specified in 4.14.6.

(b) Temperature.- Any ambient temperature in the range of -40° to +131°F.

(c) Relative humidity.- Any relative humidity up to 98 percent.

3.7.2 Equipment nonoperating.- The equipment shall meet the requirements of 3.7.1 after subjection to any one or more of the following nonoperating conditions:

(a) Temperature.- Continuous exposure for 72 hours at 160°F and 72 hours at -80°F, at any orientation.

(b) Relative humidity.- Relative humidity up to 100 percent, including condensation caused by temperature changes.

(c) Orientation.- Indefinite storage in any orientation.

(d) Vibration and bounce.- See 3.13.

3.8 Equipment shall meet the tests of Section 4.

3.9 Mechanical requirements.-

3.9.1 Worm drive gear.- Laminated construction of the worm drive gear will be acceptable.

3.9.2 The dial shall consist of the following basic components:

(1) Dial escutcheon card assembly.
(2) Finger plate.
(3) Main spring and cam assembly.
(4) Main shaft assembly with ratchet pawl.
(5) Impulse cam.
(6) Number plate.
(7) Impulse and shunt spring assembly.
(8) Governor worm assembly.
(9) Governor cup assembly.
(10) Finger stop.
(11) Main gear and ratchet assembly.
(12) Worm gear and pinion assembly.
(13) Mounting plate assembly.

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3.9.2.1 Number plate.- The number plate shall consist of a circular disk numbered and lettered on a white background. The letters shall be black in color and the numerals red on white background. The letters and numerals shall be suitably protected against abrasion resulting from the operation of the dial. The number plate shall be replaceable from the front after the finger plate is removed.

3.9.2.2 Finger plate.- The finger plate shall be of non-corrosive metal and shall be provided with ten holes .475 inch in diameter equally spaced and free from rough edges. The assembly shall be such that each hole is located over one of the numerals and the associated letters of the number plate. The finger plate shall be provided with a removable circular designation card for the station number and office name.

3.9.2.3 Finger stop.- The finger stop shall be of bright nickel or chromium finish and free from rough edges. The finger stop shall be properly secured to the mounting.

3.9.2.4 Mounting plate.- The mounting plate shall be of non-corrosive metal upon which shall be mounted the component parts marking up the complete dial assembly. The diameter of the mounting plate shall not exceed 3 inches. The overall depth of the assembled dial shall not exceed 1 5/8" and the distance between the top surface of the finger stop and bottom of the mounting plate shall not exceed 7/8". The three mounting holes in the base of the mounting plate shall be drilled and tapped for a ground head machine screw No. 4-40. The holes shall be located in accordance with the model. Three No. 4-40 mounting screws 3/8" long shall be furnished with each dial.

3.9.2.5 Impulse cam assembly.- The impulse cam shall operate the impulse spring assembly when the finger plate is returning to normal and shall open the electrical contacts of the impulse spring assembly as many times as the number of the digit dialed. The impulse cam assembly shall be so designed that each pulse will consist of a 61.5 plus or minus 2 per cent "break" and a 38.5 plus or minus 2 per cent "make" in the circuit of the impulse spring assembly.

3.9.2.6 Impulse springs.- The impulse spring and shunt spring assemblies shall be of the bifurcated type, and shall be mounted on the mounting plate by means of screws. The springs shall be readily removable for repairs or replacements without disturbing other dial component parts. The impulse springs shall be fabricated from a phosphor-bronze or nickel-silver alloy of suitable hardness and electrical conductivity, and constructed so as to cause the contacts to wipe over each other, making positive, self cleaning circuit contacts of low resistance. The contacts shall be of platinum-iridium, palladium, or other precious metal alloy highly resistant to tarnish, non-microphonic with proper adjustment, and resistant to mechanical wear. The contacts can be domeshaped or bar-shaped when the pair of contacts are arranged at right angles, and they shall be electrically welded to the springs. At any time during the normal life of the dial, the impulse spring contacts shall be capable of breaking and making a non-inductive load of 150 watts at approximately 115 volts, direct current. No evidence of excessive erosion of the metal or pitting of the contact surfaces shall occur during the normal life of the dial. Spring contacts shall not be out of alignment with respect to each other more than 1/3 of their diameter measured at the base of the contact points as gauged by eye. All springs shall be provided with screw-type terminals. The screw-type terminals shall consist of No. 4-48 machine screws with straight side binding head and shall be staked after

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assembly to prevent removal. Sufficient movement shall be provided to permit the connection of spade-type terminals. The arrangement of the screw terminals shall be such as to provide clearance between the spade clips when connected to the terminals. The separators of the spring assembly shall be of a type FBE-P-base phenolic-resin plastic complying with the requirements of Specification MIL-P-3115. The width of the separators shall be 1/16 inch greater than the width of the springs at the "pile-up" point so as to extend approximately 1/32 inch beyond the spring on each side. The impulse spring contacts shall not open when the dial finger plate is revolved to the finger stop. When the finger plate is released, the impulse cam shall open and close the impluse spring contacts the number of times corresponding to the digit dialed. When not engaged by the impulse shorting arm, the middle impulse spring should rest firmly against the heavy stop spring by its own tension. With the finger plate off normal and the tip of the main impulse spring opposite a low side of the impulse cam, the main impulse spring should rest against the contact of the middle spring, with a tension between 25 grams minimum and 50 grams maximum. With the finger plate off normal and the tip of the main impulse spring opposite a low side of the impulse cam, the heavy stop spring should hold the middle and main impulse springs, in order to make the separation between the main impulse spring and a low side of the impulse cam approximately the same as the space between contacts, when the main impulse spring is resting against the high side of the impulse cam. This adjustment is considered as having been met if, with the finger plate off normal and the tip of the main impulse spring opposite the low side of the cam, the separation between the spring tip and the cam is 0.015" + .002" and if, with the finger plate off normal and the main impulse spring resting against the high side of the cam adjacent to the locating hole, the contact separation is 0.015". All adjustments shall be made at the base of the springs and no kinks shall be apparent throughout its length.

3.9.2.7 Shunt springs.- The shunt spring assembly shall be of the bifurcated type and shall consist of five springs; three will be normally open, two normally shorted. The three normally open spring contacts of the shunt spring assembly shall close when the finger plate is operated and shall remain closed until the finger plate has returned to normal. The shunt springs shall be fabricated from a phosphor-bronze or nickel-silver alloy of suitable hardness and electrical conductivity. The contacts shall be of platinum, palladium or other precious metal alloy highly resistant to tarnish and non-microphonic with proper adjustment, dome-shaped or bar-shaped when the pair of contacts are arranged at right angles and they shall be electrically welded to the springs. At any time during the normal life of the dial, the shunt spring contacts shall be capable of breaking and making a non-inductive load of 135 watts at approximately 115 volts direct current. No evidence of excessive erosion of the metal or pitting of the contact surfaces shall occur during the normal life of the dial. The two normally closed springs will open when the finger plate is operated and will remain open until the finger plate has returned to normal. Shunt springs shall be tensioned so that, as the dial returns to normal, or moves off normal, each spring will make contact with its adjacent spring while traveling through a space of not less than 0.015". The contact separation for shunt springs in their normal position shall be from 0.015" and to 0.030". When the dial is in off-normal position, contact separation shall be maximum 0.050". The shunt spring operating cam shall be aligned within the width of the buffer on the operating shunt spring in the normal position. All adjustments shall be made at the base of the springs and no kinks shall be apparent throughout its length.

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3.9.2.8 Impulse shorting arm.- The impulse shorting arm shall cause the main impulse spring to clear the cam by a minimum of 0.015", maximum 0.030" during the shorted impulse. The impulse shorting arm shall not cause the impulse springs to short until after the completion of the last transmitted pulse dialed and the impulse springs shall remain shorted until the beginning of the next transmitted pulse. A plane parallel to the impulse spring, shall pass through a point within the width of the contracting portion of the impulse shorting arm.

3.9.2.9 Governor assembly.- The governor assembly shall consist of an adjustable device to control the speed at which the finger plate returns to its normal position after being operated, and consequently the speed of operation of the impulse springs. The governor wings shall be formed as nearly as possible alike. The worm shall be made of stainless steel.

3.9.2.10 Ratchet pawl.- With the dial at normal, the pawl shall rest against its stop so as to give minimum 0.008", maximum 0.030" clearance between the shaft stop and its associated stop. Action of the ratchet mechanism shall be reasonably silent.

3.9.2.11 Drive assembly.- The drive assembly shall consist of the arrangement of springs, gears, cams, ratchets, shafting and other necessary mechanism to operate the governor, the pulsing spring assembly and the shunt spring assembly in the manner previously described in this specification. Means shall be provided for readily adjusting the tension of the main drive spring. This restoring spring shall have one to one and a third turns tension with the dial at normal. End play in the worm gear shaft shall not cause all of the active face of the worm gear to move out of intersection with the horizontal center line of the plane of the worm.

3.9.2.12 Lubrication of parts.- Each of the parts listed below shall be lubricated with one dip of watch oil, per Specification MIL-L-3918, using No. 4 artist's sable rigger brush:

(a) Dial shaft bearings.

(b) Main gear wheel bearing.

(c) Exposed portion of main bearing on governor side of mounting plate.

(d) Governor shaft worm.

Each of the parts listed below shall be lubricated with one drop of watch oil, per Specification MIL-L-3918:

(a) Worm gear shaft bearings.

(b) Governor shaft bearings.

(c) Pawl bearing.

Each of the parts listed below shall be lubricated with one dip of light mineral oil, Specification VV-L-820. Excessive oil should not be allowed to remain on any surface.

(a) Ratchet teeth on main gear.

(b) Cam edge.

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(For the purpose of this specification: a dip shall be considered to be the amount of lubricant retained on the brush after the brush has been dipped into the lubricant to a depth of 3/8 inch and then scraped on the edge of the container to remove the surplus oil. There should not be sufficient oil adhering to the brush to form a drop at the end of the bristles; a drop shall be considered the amount of oil released from a piece of number 22 B & S gauge, bare tinned copper wire after it has been dipped ½ inch into the lubricant and quickly withdrawn).

3.9.2.13 Dust shield.- A paper dust shield shall be placed between the dial mounting and the number plate to prevent dust from entering the drive assembly.

3.9.3 Fit.- The dial shall be capable of fitting into, and operating properly in the following telephones:

A. Signal Corps Telephone TP-6A as made by Connecticut Telephone and Electric Corporation and North Electric Company.

B. AN Telephone TA-166/U as made by Connecticut Telephone and Electric Corporation.

C. Western Electric Co., Telephone No. 302AW.

D. Automatic Electric Co., Telephone No. 40.

E. Kellogg Switchboard and Supply Co., Telephone No. 925BAX.

F. Stromberg Carlson, Telephone No. 1222 BW and 1242 WA.

G. North Electric Manufacturing Co., Telephones Nos. 3H68L and H6008L.

3.10 Electrical requirements.-

3.10.1 Impulse interval.- Impulse mechanism shall be capable of providing an elapsed time interval of approximately 125 milliseconds between the completion of the last pulse of a digit and the return of the finger plate to its normal position.

3.10.2 Dial speed (pulses per second).- The dial shall be so adjusted as to provide pulses at the rate of 10 per second when tested in accordance with the procedure described in 4.12. A variation of ± 1 pulse per second is permissible.

3.10.3 Percentage break.- The dial shall be so designed that each pulse shall consist of a 61.5 pulse or minus 2 percent "break" and a 38.5 plus or minus 2 percent "make" in the electrical circuit at the impulse spring contacts when tested in accordance with the procedure specified in 4.13.

3.11 Life requirement.- The dial shall be capable of at least 500,000 dialing operations. The dial shall be capable of meeting all the electrical requirements of this specification at the conclusion of 500,000 dialing operations. One dialing operation shall consist of dialing the number zero and release for return to normal position.

3.12 Breakdown voltage.- Air gaps and electrical insulation shall have a minimum breakdown voltage of 500 volts when tested in accordance with 4.10.

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3.13 Vibration, bounce and shock.-

3.13.1 Vibration.- Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed.

3.13.2 Bounce and shock.- The equipment shall meet the requirements of table I. See paragraphs 4.14.1 and 4.15.

Table I - Bounce and shock

Requirement	Inspection Paragraph	Performance after inspection
Bounce; light equipment	4.14.1	Full specification performance. No physical damage except surface abrasion.
Shock; drop	4.15	Equipment shall be operable. Minor mechanical damage is permissible.

3.14 Preconditioning.- The equipment shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.

3.15 Preproduction samples.- The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract. (See 6.2(e)).

3.16 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2(h)).

3.17 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein.

3.2.2 Flux and cleaning agents for soldering.
3.2.3 Process for soldering.
3.3 Cleaning of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2).

(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3 and 4.5).

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4.1 Classification of inspection.- (Cont'd)

(1) Procurement inspection of items before preparation for delivery. (See 4.3).

(2) Procurement inspection of preparation for delivery. (See 4.5 and section 5).

4.2 Preproduction inspection.- This inspection shall consist of the preproduction inspection specified in table II and the group A and group C inspection specified in tables III and IV, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, ballistic or drop, and (5) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II - Preproduction inspection

Inspection (For additional preproduction inspection see 4.2)	Reqt Par.	Test Par.
Temperature	3.7.1	4.14.3
Moisture Resistance	3.7.1	4.14
Bounce	3.13	4.14.1
Drop	3.13	4.15
Life	3.11	4.14.6
Vibration	3.15	4.14.2
Interchangeability	3.4	4.7

4.3 Procurement inspection of items before preparation for delivery.- The minimum inspection performed by the contractor shall be as specified by (a) through (c) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.6). Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A and group C inspection as specified in 4.3.1 through 4.3.3.2.

(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

(c) Each unit which will be subjected to group A or group C inspection shall be preconditioned after final assembly. (See 3.14).

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4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.14.5) shall be last. (See 6.7).

Table III - Group A inspection

Inspection	Reqt. Par.	Insp. Par.	AQL Major	AQL Minor
Visual & Mechanical				
Workmanship	3.17		1.0%	4.0%
Impulse cam	3.9.2.6	4.11		
Electrical				
Dielectric	3.12	4.10		
Speed	3.10.2	4.14.5.2	1.0%	*
Make and break	3.10.3	4.13		
*All electrical defects are of major category.				

4.3.2 Group B inspection.- Not required.

4.3.3 Group C inspection.- This inspection shall be as listed in table IV.

4.3.3.1 Sampling for inspection of equipment.- One equipment for 1000 units or fraction thereof produced on contract or order shall be selected for each group C inspection.

4.3.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production of the items on contract.

4.3.3.3 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4).

Table IV - Group C inspection

Inspection	Reqt. Par.	Insp. Par.
Fit	3.9.3	4.9
Accelerated life	3.11	4.14.6

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4.3.3.4 Disposition of samples.- Samples which have been subjected to the accelerated life test of group C inspection shall not be furnished on contract or order.

4.4 Bounce preconditioning.- The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to procurement inspection.

4.5 Procurement inspection of preparation for delivery.- The minimum inspection performed by the contractor shall be as specified in 4.5.1 and 4.5.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government--at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.6). Such verification shall not exceed the amount of inspection required herein of the contractor. When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.5.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table V.

TABLE V

PRESERVATION AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Wrong type of contact preservative.
2. Quantity in unit package not as specified.	
3. Incorrect packaging method applied.	2. Item not properly blocked or braced within the unit package to prevent movement.
4. Item not properly cleaned prior to preservation.	
5. Item not protected with contact preservative.	3. Packaging material damaged.
6. Incomplete or improperly applied contact preservative, such as insufficient overlap, air bubbles, etc.	4. Conforming wraps are not snug-fitting and contain voids.
7. Grease proof wrap omitted.	5. Any item of marking information other than 12 through 14 omitted, incorrect, or illegible.
8. Cushioning or padding omitted.	
9. Cushioning or padding inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item.	
10. Cushioning inadequate for the physical and mechanical protection of the item.	
11. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
12. Stock number omitted, incorrect, or illegible.	
13. Nomenclature omitted, incorrect, or illegible.	
14. Marking of quantity of items in package omitted, incorrect, or illegible.	
15. Different stock numbered items in the same package.	

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4.5.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VI.

TABLE VI

PACKING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Gross weight in excess of specified amount.	2. Defective taping or sealing of carton.
3. Box closure not as specified.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Type, grade, class, and style of the shipping container not as specified.	4. Any item of required marking information other than 8 through 14 listed under major defects omitted, incorrect, or illegible.
5. Strapping omitted.	
6. Strapping inadequate or incorrectly applied.	
7. Shipping documents or packing list omitted.	
8. Stock number omitted, incorrect, or illegible.	
9. Nomenclature omitted, incorrect, or illegible.	
10. Marking of quantity of items in pack omitted, incorrect, or illegible.	
11. Destination marking omitted, incorrect, or illegible.	
12. Service designation (color marking) omitted.	
13. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
14. Overseas code marking omitted, incorrect, or illegible.	

4.5.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.5.2.2 Procedure in case of failure.- If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.5.2.3 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

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4.6 Rough handling test (preparation for delivery).- When rough handling test is required by the contract (see 6.2(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling.

4.7 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

4.8 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VII.

Table VII - Classification of Visual and Mechanical Defects

Classification	Defect
Major	Improper lead dress or cabling or wires. Insulation burned, abrased, pinched. Taut wire. Insulation frayed.
Minor	Broken strands. Insulation back more than 1/8 inch.

4.9 Fit test.- The dials produced on the order shall be tested to determine if they fit and operate properly in each of the telephones mentioned in 3.9.3. Failure of the dial to fit and operate properly in any one of the telephones mentioned in 3.9.3 shall be cause for rejection of the dial.

4.10 Dielectric test.- Insulation in the assembled dial shall be tested for compliance with 3.12 by an alternating current potential of 500 volts r.m.s. applied for a period of 30 seconds between conductors and any insulated electrical part.

4.11 Mechanical operation test.- Operate the dial finger plate from normal position by placing the index finger in the finger plate opening over the digit "0" and revolving the finger plate in a clockwise direction until the finger is against the finger stop. Release the finger plate by removing the index finger, and note that the finger plate returns to normal with a smooth and uniform motion. Note that the impulse cam does not open the impulse spring contacts while the finger plate is being operated. Note that the shunt spring assembly operates when the finger plate is operated and remains operated when the finger plate is returning to normal until after the last pulse.

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4.12 Speed tests.- Connect the dial under test to dial speed test set approved by the contracting officer. Operate and release the dial finger plate as described in 4.11. A dial speed of 10 pulses per second ± one pulse shall be indicated on the dial speed test set.

4.13 Percentage make and break test.- Connect the dial under test to a dial "percentage make and break" test set approved by the contracting officer. Operate and release the dial finger plate as described in 4.11. A percentage of 61.5 plus or minus 2 percent "break" and a percentage of 38.5 plus or minus 2 percent "make" shall be indicated on the test set.

4.14 Service conditions tests.- The following service conditions tests shall be performed on the pre-production samples and on samples selected for Group C inspection as indicated in table IV. During the service conditions tests, the equipment shall be operated continuously or intermittently as the specification requires, and performance data recorded. Between tests the equipment shall be checked for such defects as low dielectric strength, loss of electrical continuity, loose parts, excessive corrosion, breakage, etc. Performance checks shall be made on the equipment at the conclusion of each service condition test. These performance checks shall be sufficiently adequate in number and scope, to uncover any defect resulting from the preceding service conditions tests. Any evidence of degradation of performance below limits specified herein shall be cause for rejection.

4.14.1 Bounce test; equipment under 1000 pounds.- The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ±1 percent, for a total of 3 hours. During each ½ hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.14.2 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.13.1, as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.14.2(d)).

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

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4.14.2 Vibration test.- (Cont'd)

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.14.3 Temperature.- The dials shall be temperature cycled in accordance with MIL-STD-169. Test shall be made in accordance with 4.12 during steps 4, 8, and 10. During step 8 the dial shall make not less than 8, nor more than 12 pulses per second when dialed in the normal manner. After step 10, the equipment shall be tested in accordance with 4.12 and 4.13.

4.14.4 Moisture-resistance test.- Dials shall meet the following moisture-resistance test:

4.14.4.1 Test procedure.- The test procedure shall be as follows:

(a) Dry at 130° ± 5°F for 24 hours.

(b) Test dial for operation and speed in accordance with 4.11 and 4.12.

(c) Subject to continuous cycling for a minimum of 5 of the 48-hour cycles shown on MIL-STD-170, and check the mechanical operation and dial speed in accordance with 4.11 and 4.12, and insulation resistance in accordance with 4.14.5.3 during periods indicated on that drawing.

(d) Do not remove from test chamber for tests.

(e) Start measurements not more than 5 minutes after power has been applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on the equipment after measurements have been completed.

4.14.5 Performance requirements.-

4.14.5.1 Mechanical operation.- The mechanical operation shall meet the requirements in 4.11 at any time during the cycling test period.

4.14.5.2 Dial speed requirements.- The dial speed shall not at any time during the cycling test period be less than 8 or greater than 12 pulses per second when tested in the manner described in 4.12.

4.14.5.3 Insulation requirements.- The insulation resistance between shunt spring terminals and between shunt spring terminals and the mounting plate, with the finger plate in normal position, and the resistance between pulse spring to terminals and between pulse spring terminals and mounting plate with the finger plate operated so that the pulsing springs are open shall not be less than 1 megohm when measured with a d.c. potential of not less than 100 volts applied for one minute.

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4.14.5.4 Failure.- If any component fails to meet the specified performance during cycling, it shall be realined or readjusted once. If the component then fails to meet specified performance or fails subsequently during cycling, it does not pass the test. In addition, if the component fails to meet full specification requirements, it shall be considered as failing to pass the test.

4.14.6 Accelerated life test.- Dials shall be life tested by a dial testing machine approved by the contracting officer which simulates a telephone instrument and central office equipment. The machine shall stop automatically if the dial pulsing speed falls below 8 pulses per second or if the dial fails to make the correct number of pulses. The dials shall operated satisfactorily for at least 500,000 dialing operations, and at the conclusion of that number of operations shall meet the requirements of 4.11, 4.12 and 4.13.

4.15 Shock test; drop.- The equipment, in its transit c. combination case, shall be subjected to the drop test in accordance with table VIII. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

Table VIII - Drop test

Weight of equipment including case	Largest dimension of case (inches)	Notes	Height of drop (inches)	Number of drops
Under 100 pounds	36 and over	A	30	Drop on each face, edge, and corner. Total of 26 drops.

Note A: Drops shall be made from a quick-release hook.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Dial Telephone (Dial TA-45( )/TG) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Dial TA-45( )/GT.-

5.1.1.2 Cleaning.- The process of cleaning shall be at the option of the contractor.

5.1.1.3 Drying.- Drying process shall be at the option of the contractor.

5.1.1.4 Preservation.- Preservative application shall be at the option of the contractor.

5.1.1.5 Unit packaging.- Following the preservation application each dial shall be packaged Method IA-8 as follows: Each dial shall be wrapped in greaseproof barrier material type II, grade A, class 1 conforming to Specification MIL-B-121. Insert the wrapped dial within a close-fitting bag made of water-vaporproof barrier material qualified for class 2 application conforming to Specification MIL-B-131 and fabricated in accordance with Specification MIL-B-117. With the dial inserted partially deflate the bag by careful hand working or by an exhaust system, to a minimum volume without endangering the bag by puncture. Heat seal the bag to effect closure.

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5.1.1.6 Intermediate packaging.- A quantity of unit packages shall be placed within a close-fitting regular slotted style fiber box, class 2 W6c conforming to Specification PPP-B-636. Weight of box and contents shall not exceed 30 pounds. Box closure shall be effected by the application to all seams and joints of pressure-sensitive water resistant tape conforming to Specification PPP-T-76.

5.1.2 Level C.- Dial Telephone (Dial TA-45( )/GT) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- A quantity of intermediate packages (see 5.1.1.6) shall be packed for shipment within close-fitting overseas class or type wirebound wood boxes, fiberboard, wood cleated boxes, wood, cleated-plywood boxes, or nailed and lock-corner boxes, conforming to PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621 as applicable. The weight of any one box with contents shall not exceed 200 pounds. Box closure shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in accordance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Dial Telephone (Dial TA-45( )/GT) shall be packed for shipment as specified in 5.2.1 except shipping containers shall be domestic type or class.

5.2.3 Level C.- Dial Telephone (Dial TA-45( )/GT) shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4, and Consolidated Freight Classification, Rule 21, for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment. Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Dial TA-45( )/GT is used with telephones of various types to secure automatic selection of individual telephone stations, when used with automatic telephone central office equipment.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.
(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.15).

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6.2 Ordering data.- (Cont'd)

(2) .....
(3) Preproduction pack(s) as follows:

a. Makeup of pack(s).
b. Number of each kind of pack to be submitted.
c. Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.
d. Five copies of process sheet, using format shown on Drawing Number SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack.
e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required. (See 3.16)
(i) Quantity of tools and running spare parts required. (See 3.16)
(j) Submission of the statement of treatment referenced in 3.6.2.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: TA-45W/GT. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information. (See 1.1).

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.3).

6.5 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.5.1 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.5.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.6 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

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6.6 Verification inspection.- (Cont'd)

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

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